DATA ANALYTICS / PORTFOLIO

SUPPLY CHAIN ANALYTICS

Fulfillment, service reliability & profitability

Power BI · DAX · Operational Analytics

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FROM ORDER TO DELIVERY



A decision-focused portfolio case study

Project objectives

An interactive decision-support dashboard, not just a set of charts.

01 / COMMERCIAL PERFORMANCE

Track value creation

Monitor sales, profit, order volume and monthly patterns across business segments.

02 / DELIVERY RELIABILITY

Diagnose late shipments

Compare shipment performance by region and shipping mode to surface exceptions.

03 / PROFITABILITY

Expose loss-making activity

Identify unprofitable orders and categories that deserve closer investigation.

04 / ACTION

Turn insights into priorities

Enable operations and finance teams to focus on a manageable review queue.

The operational question

How can a business protect profit when delivery reliability is under pressure?

Sales alone cannot reveal where shipping delays and loss-making orders coexist.

LATE DELIVERY RATE

57.2%

Reported on the dashboard with all visible slicers set to “All”.

Business question → Where should we investigate first?

PROJECT SNAPSHOT

Commercial scale & service context

Dashboard-reported values from the supplied Executive Overview.

TOTAL SALES

$2.25M

TOTAL PROFIT

$248.63K

TOTAL ORDERS

17,350

CUSTOMERS

9,300

57.2% Late delivery rate

3.5 days Average shipping days

THE APPROACH

From transactions to operational intelligence

A reproducible analytics workflow linking commercial results to fulfillment exceptions.

01

TRANSACTION DATA

Orders, products, customers, shipping

02

DATA PREPARATION

Clean fields, align dates,
validate IDs and grain

03

ANALYTICAL MODEL

Order and item context, date and category filters

04

DAX & KPI LOGIC

Sales, profit, late rate, shipping-time measures

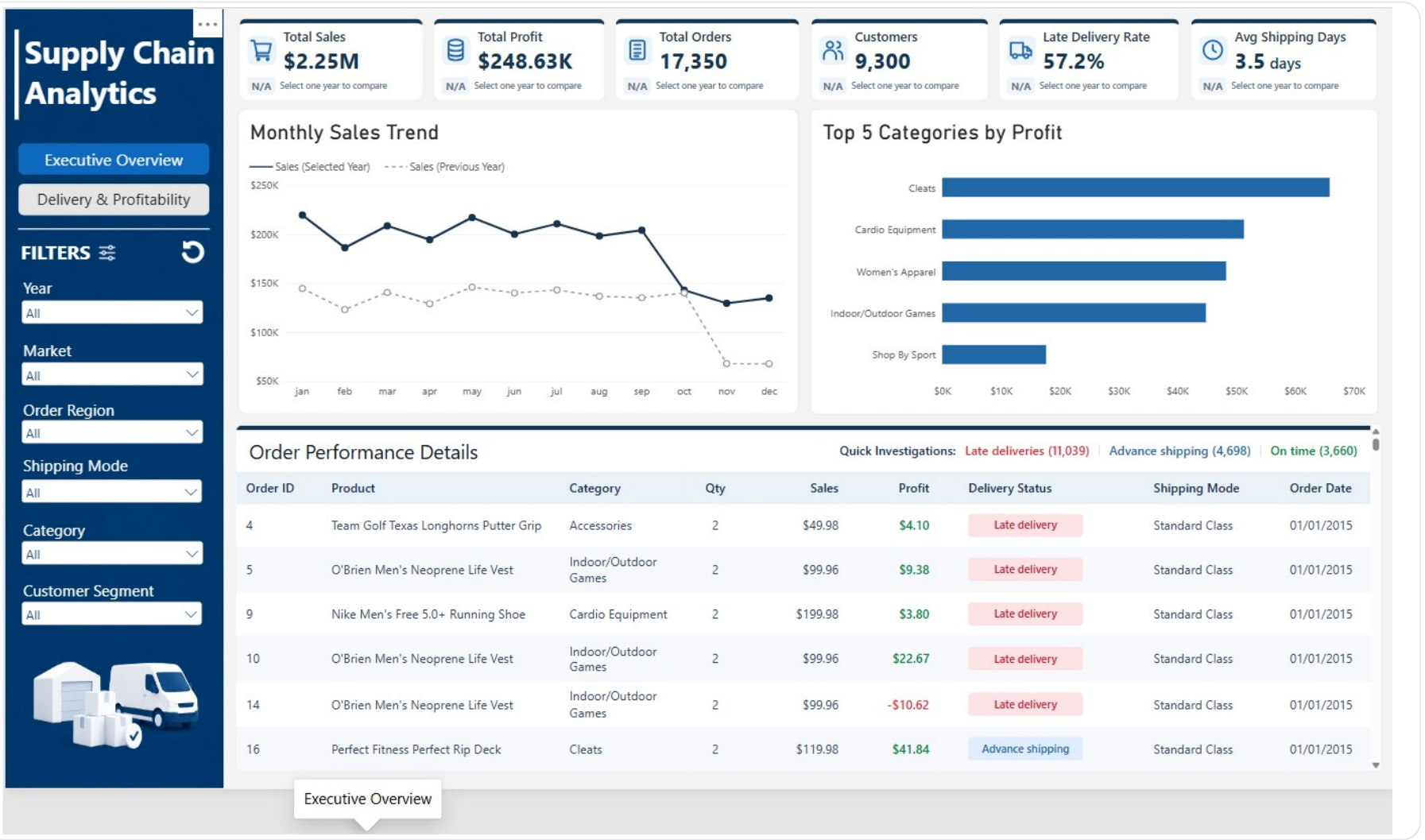
05

DECISION SUPPORT

Interactive views and order-level investigation

MEASUREMENT DISCIPLINE Validate order grain, delivery definitions and profit aggregation before treating KPIs as final.

A commercial command center



01 / VALUE

Sales & profit

Compare commercial performance.

02 / PATTERNS

Monthly trends

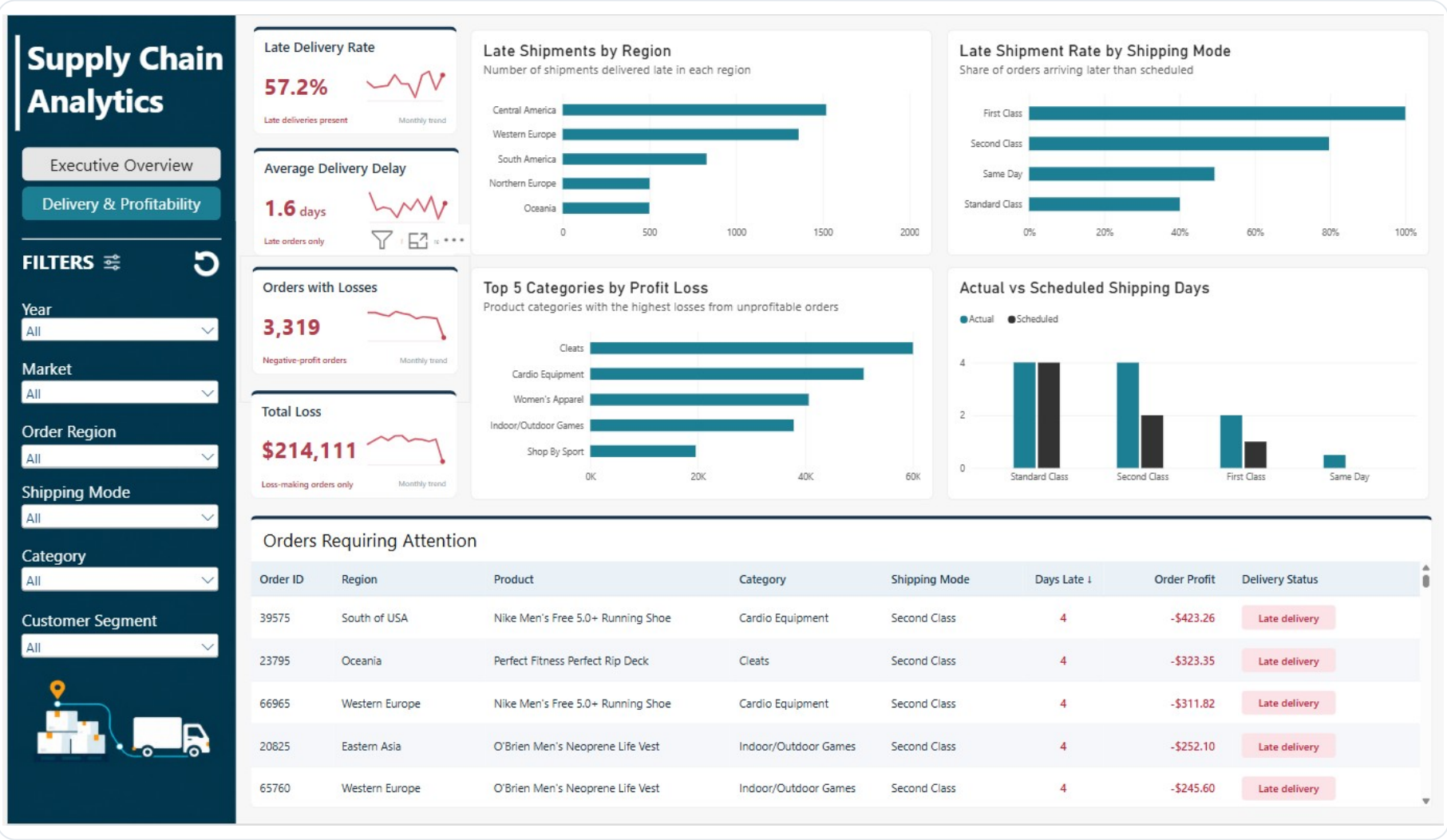
Monitor variation across the year.

03 / DETAIL

Order drilldown

Investigate product-level records.

The operational diagnostic layer



01 / WHERE

Regional delays

Locate concentrated late-shipment counts.

02 / HOW

Shipping modes

Compare service performance.

03 / IMPACT

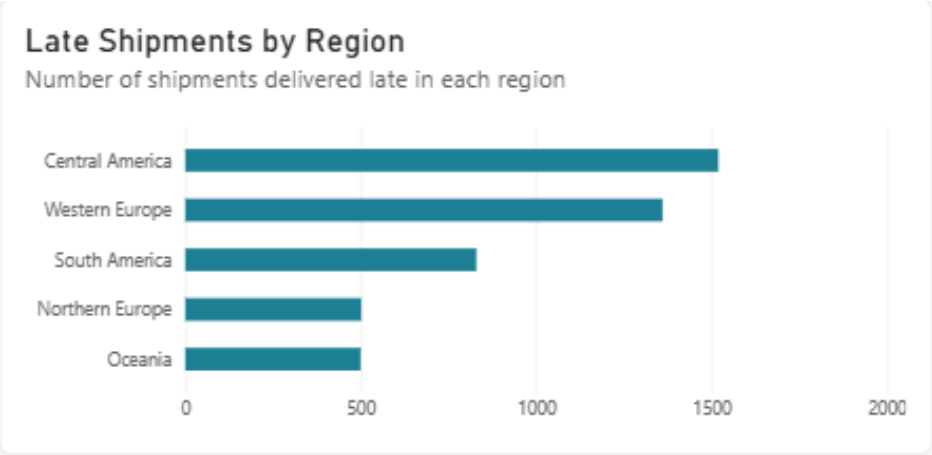
Loss-making orders

Bring anomalies to order-level detail.

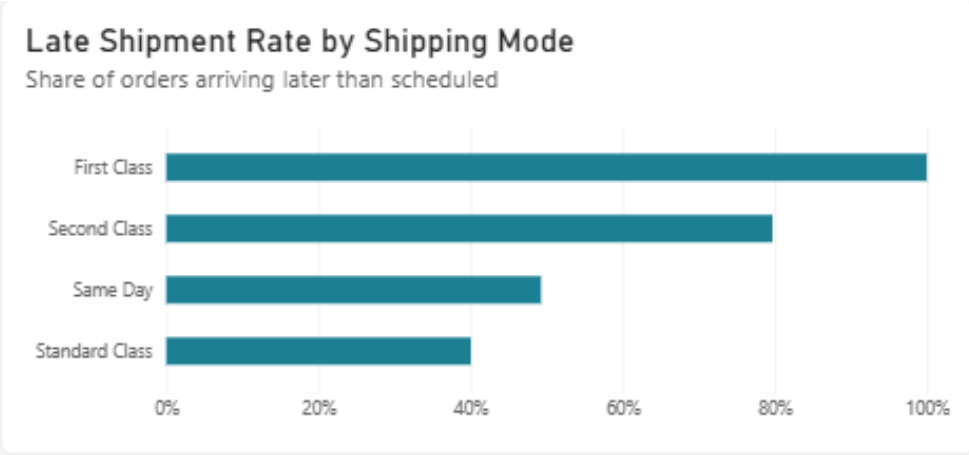
Two lenses on late shipments

Volume and rate serve different purposes; a high count is not the same as a high percentage.

LATE SHIPMENTS BY REGION



LATE RATE BY SHIPPING MODE



CONCENTRATION

Central America and Western Europe lead the visible regional count ranking.

RATE

First Class has the highest displayed late-rate bar; audit cohort definitions before action.

Loss-making activity needs a separate lens

The dashboard shows loss exposure and fulfillment exceptions; it does not prove causation.

ORDERS WITH LOSSES

3,319

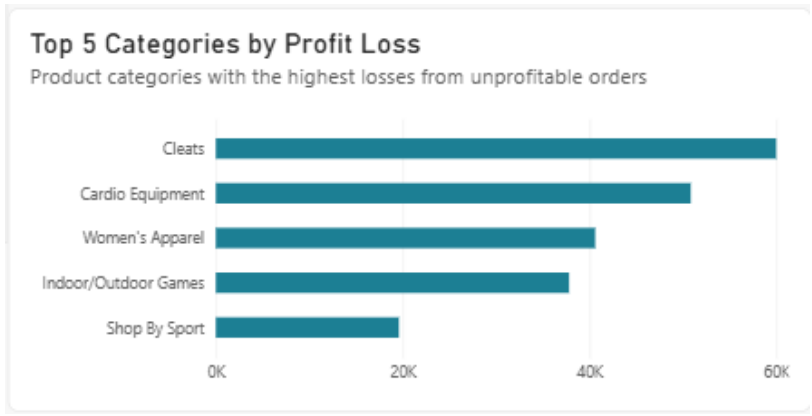
TOTAL LOSS

$214,111

AVG. DELIVERY DELAY

1.6 days

TOP 5 CATEGORIES BY PROFIT LOSS



READ THE SIGNAL CAREFULLY

Cleats leads the displayed loss-category ranking.

This is a diagnostic priority, not evidence that late delivery caused the loss.

CASE INVESTIGATION

From an alert to an order-level review

The “Orders Requiring Attention” table connects operational exceptions to specific orders.

01 FILTER
Select region or mode

02 IDENTIFY
Sort late orders

03 INSPECT
Review days late & profit

04 ESCALATE
Validate causes with owners

Orders Requiring Attention

Order ID	Region	Product	Category	Shipping Mode	Days Late ↓	Order Profit	Delivery Status
39575	South of USA	Nike Men's Free 5.0+ Running Shoe	Cardio Equipment	Second Class	4	-$423.26	Late delivery
23795	Oceania	Perfect Fitness Perfect Rip Deck	Cleats	Second Class	4	-$323.35	Late delivery
66965	Western Europe	Nike Men's Free 5.0+ Running Shoe	Cardio Equipment	Second Class	4	-$311.82	Late delivery
20825	Eastern Asia	O'Brien Men's Neoprene Life Vest	Indoor/Outdoor Games	Second Class	4	-$252.10	Late delivery
65760	Western Europe	O'Brien Men's Neoprene Life Vest	Indoor/Outdoor Games	Second Class	4	-$245.60	Late delivery

EXAMPLE ON SCREEN Order 39575 · 4 days late · -$423.26 order profit · South of USA

FINDINGS

What the dashboard brings to the surface

SERVICE

57.2%

Displayed late delivery rate

GEOGRAPHY

2 regions

Central America and Western Europe top late-shipment counts

SHIPPING

First Class

Highest displayed shipping-mode late rate

CATEGORY

Cleats

Visible leader in profit and gross-loss category rankings

THE MANAGEMENT IMPLICATION

Prioritize where service risk, order scale and profitability intersect.

Business recommendations

Concrete investigation steps, based on signals from the displayed dashboard.

01

Prioritize late-delivery hotspots

Review Central America and Western Europe by order volume, date and shipping mode.

OPERATIONS

02

Audit the shipping-mode comparison

Validate First Class and other service-level definitions and denominator logic.

OPERATIONS + BI

03

Investigate unprofitable orders

Segment negative-profit orders by category, shipping service and commercial terms.

FINANCE + CATEGORY

04

Track change over time

Set agreed definitions, service targets and an ongoing exception-review cadence.

OPERATIONS + BI

DATA INTEGRITY

A good analyst validates the story

An explicit reconciliation checkpoint before presenting the metrics as production-validated.

DASHBOARD SNAPSHOT

17,350

orders reported in the Power BI visual

SEPARATELY SUPPLIED CSV

15,549

rows across 41 columns in the extract

RECONCILIATION CHECKLIST

- Confirm that CSV and PBIX use the same population and refresh.
- Validate order IDs, date consistency and order/item grain.
- Recompute timing and profit measures from approved business definitions.

The supplied CSV alone does not contain the actual vs. scheduled shipping-day fields shown in the report.

CAPABILITIES

Skills demonstrated

Data preparation → Model → Measure → Visualize → Decide

POWER QUERY

Preparation & validation

Source cleaning, data types and consistency checks.

DATA MODELING

Analytical structure

Order, product, category, customer and time context.

DAX

Business metrics

KPI logic, conditional context and delivery calculations.

POWER BI

Interactive reporting

Multi-page dashboards, slicers and cross-analysis.

REPORT DESIGN

Usable investigation

Custom visual hierarchy, status flags and row-level review.

DATA STORYTELLING

Decision support

Translating findings into operational review priorities.

THANK YOU

From order data to better delivery decisions.